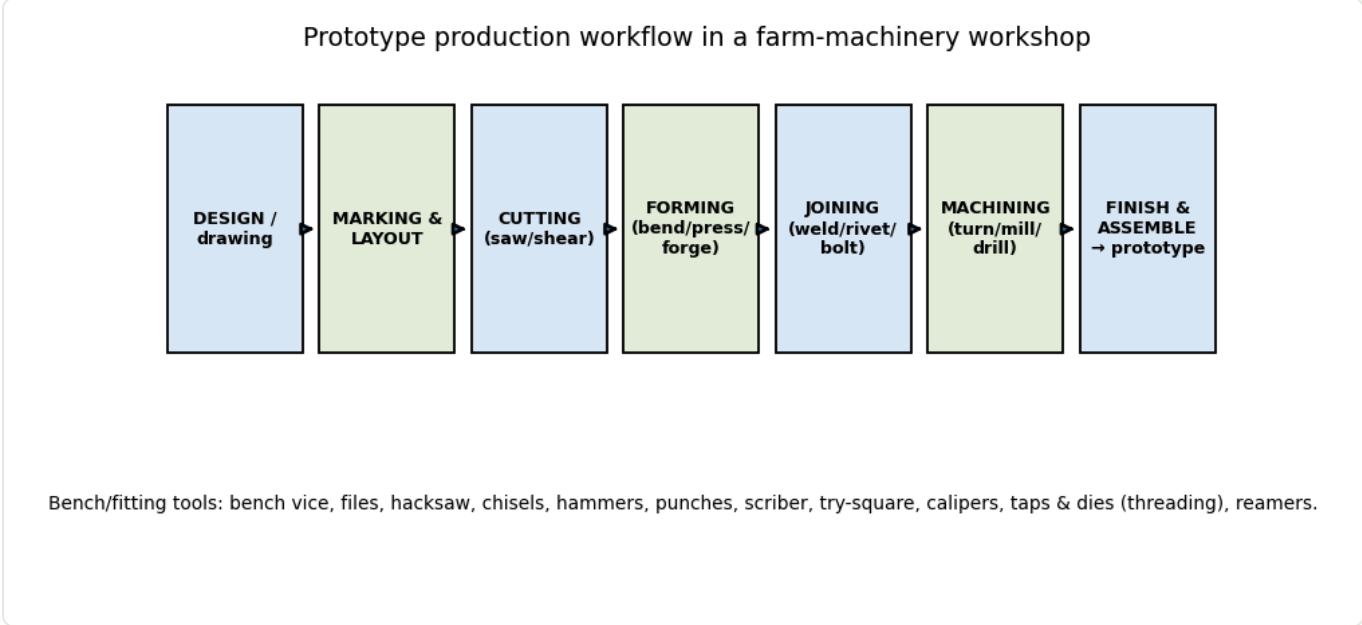
UNIT 8 · TOPIC 2

Workshop Practice in Prototype Production

Syllabus topic: Workshop practices applied in prototype production



Prototype production workflow in a farm-machinery workshop

Steps in making a farm-machine prototype

1. Design / drawing.
2. **MARKING & layout** — scriber, punch, try-square, surface plate.
3. **CUTTING** — hacksaw, shear, gas/plasma.
4. **FORMING** — bending, pressing, forging, rolling.
5. **JOINING** — welding, riveting, bolting (Topic 6a).
6. **MACHINING** — turning, milling, drilling to final size (Topic 4a).
7. **FINISHING & ASSEMBLY**, then testing (Unit 3).

- Prototype work is largely MANUAL and one-off, so it relies on skilled fitting and general-purpose machines.
- Batch/mass production later adds jigs, fixtures, dies and dedicated tooling (Topics 3 and 4b).

Practice MCQs

Q36. The correct order of prototype fabrication is:

(a) Machining → design → cutting

(b) Design → marking → cutting → forming → joining → machining → assembly

(c) Assembly first

(d) Welding → design

Answers

Q36 — (b) Design → marking → cutting → forming → joining → machining → assembly

Sources: Hajra Choudhury — Elements of Workshop Technology (Vols I & II); Kalpakjian — Manufacturing Engineering & Technology; standard production-engineering texts; NPTEL manufacturing lectures.
